RUNI – Algorithms – Homework 2

Due date: Sunday 16/11/2025

NOTE: the general homework guidelines published in homework 1
are valid for all homework assignments in this course.

Problem 1:

a. Let G=(V,E) be a **connected undirected** graph.

Prove that it is possible to add at most $\lfloor n/2 \rfloor$ edges to E such that the resulting graph has an Euler cycle.

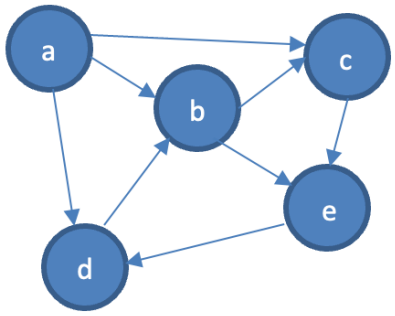
b. Let G=(V,E) be a **directed** graph such that for every $v \in V$, $d_{in}(v) = d_{out}(v)$

Prove that G is strongly connected if and only if its underlying graph is connected.

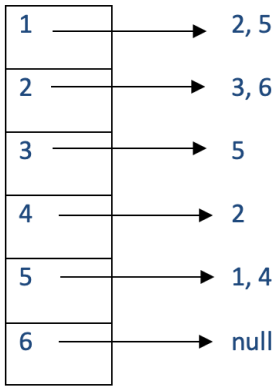
Problem 2:

For each of the following graphs find and present (separately) an Euler path and a Hamilton path, or claim that they do not exist (no proof required).

a. The graph provided in the following figure:

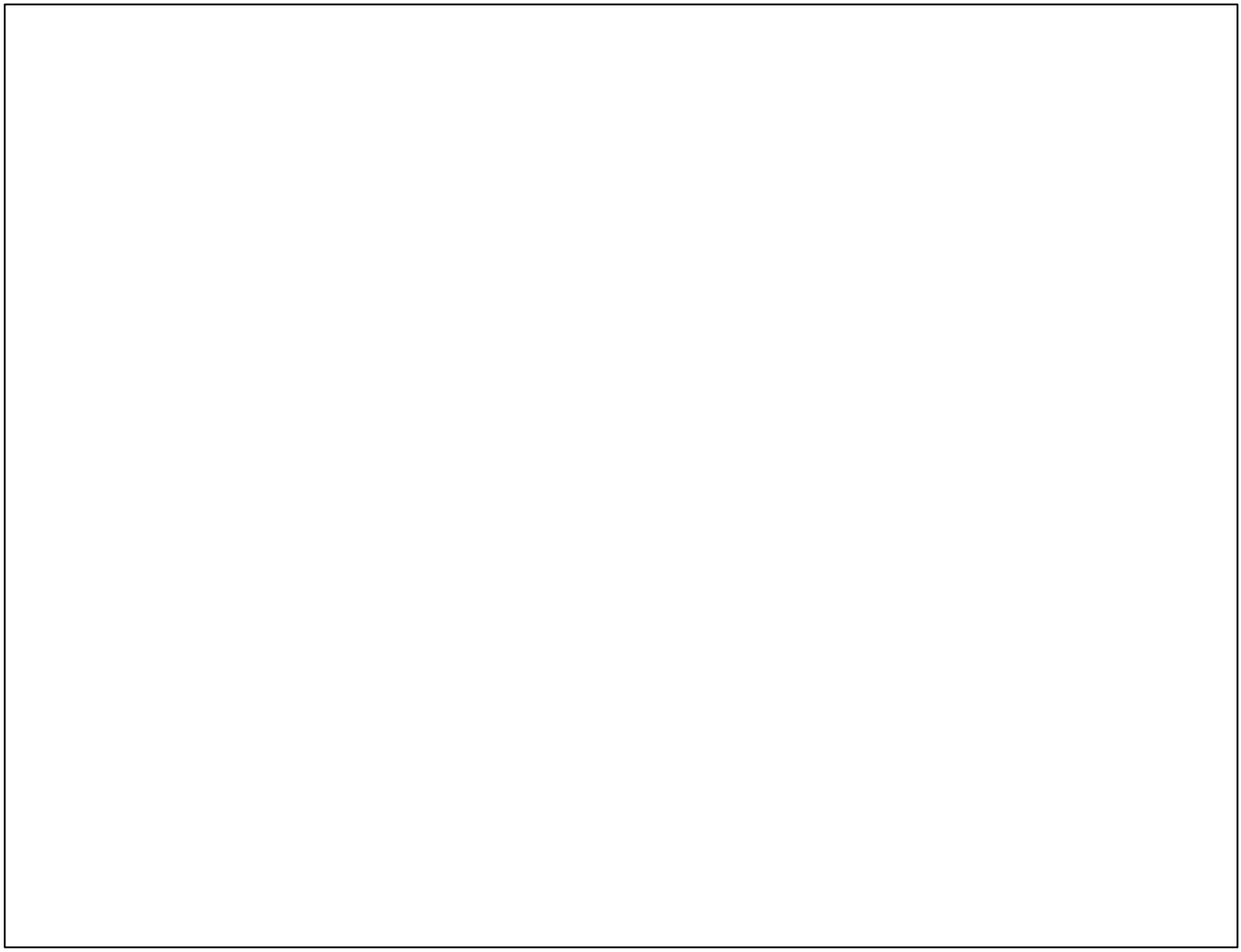


b. The graph provided in the following representation:



c. The graph provided in the following representation:

	a	b	c	d	e	f
a	0	1	0	0	0	0
b	0	0	1	0	1	1
c	0	0	0	0	1	1
d	0	0	1	0	0	0
e	0	2	0	0	0	0
f	0	0	0	1	0	0



Problem 3:

In each of the following questions, select the correct answer and justify. No need to refute the wrong answers.

3.1 The sequence 00120221 is a De-Bruijn **sub**sequence over $\Sigma = \{0,1,2\}$ and some n.

a. It must be that $n = 2$.
b. If the next character is 0, then it must be that $n > 2$.
c. There is no Euler cycle in the De-Bruijn graph $G_{3,2}$.

3.2 Let G be a graph in which every Hamiltonian cycle is also an Euler cycle.

a. G must be a clique

b. G must be a cycle

c. G is not necessarily a clique or a cycle.

3.3 Let $G = (V_1, V_2, E)$ be an **undirected bipartite** graph **with an Euler cycle**.

a. $|V_1| = |V_2|$
b. $|V_1|$ and $|V_2|$ are even.
c. $\sum_{u \in V_1} \deg(u) = \sum_{v \in V_2} \deg(v)$

Problem 4:

Let G=(V,E) be an **undirected connected** graph, such that **all its vertices have even degrees**.

A vertex *s* is called an Euler-center if every application of the algorithm for finding an Euler cycle in G, starting at *s*, produces an Euler cycle in a single iteration, no matter what is the order in which edges are selected. In other words, there is no need to merge different parts of the cycle into others. Complete the proof of the following theorem:

If *s* is an Euler-center then *s* appears in every simple cycle in G.

Proof: Assume that *s* is an Euler-center.

Assume towards contradiction that s does not appear in some cycle C. Let G\C be the graph G with the edges of C removed. Formally, V(G\C)=V, E(G\C)=E\C.

(complete the proof)

Problem 5:

In a recent debacle, thieves disguised as construction workers stole eight pieces of the French Crown Jewels valued at approximately €88 million from the Galerie d'Apollon of the Louvre in Paris, France. Lucas Philippe, head of the French interpol requested your help in analyzing the patrol route of the night of the incident.

The head guard at the Louvre shared the following details:

- There are seven major areas: Apollo (A), Denon (D), Egyptian (E), Greek (G), Napoléon (N), Richelieu (R), and Sully (S).
- There is one bidirectional (undirected) corridor between every pair of areas.
- A single guard performs a continuous patrol walking each corridor exactly once.

a. Will a patrol route that uses every corridor exactly once and returns to its starting area be possible under these conditions? Give a brief yes/no justification.

b. On occasion, corridors may close for renovation. Given a subset E' of open corridors, propose a method to efficiently decide whether a continous patrol that uses all corridors exactly once exists.

c. After much investigation, it was found that on the day of grand burglary the area Sully (S) was entirely closed, along with all corridors connected to it. The night guard reports he managed to adjust his patrol and had performed a route that uses every corridor exactly once, and starts and ends in the Apollo area. Is the guard telling the truth? Give a brief yes/no justification.